

Flow properties of coarse and fine sugar powders

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Abstract

The densification dynamics and flow properties of powdery to granular varieties of commercial sugars were obtained. Samples of granular sugar (GS), refined (RS) sugar, a raw type VHP sugar (VS), and icing sugar (IS) were characterized concerning particle morphology, size distribution, loose and tapped bulk densities. Primary particles of IS sugar have a mean size (D_{50}) of 77 μm and they agglomerate into larger size granules with intraparticle voidage. The RS, GS and VS sugars have D_{50} equal to 286 μm , 537 μm , and 696 μm , respectively. The RS sugar revealed some irregular and rough agglomerates, while the other samples were composed of flat and well-defined crystals. After full compaction, the initial bulk volume of the IS sugar was reduced by 25%, of the RS sugar by 18%, of the GS and VS sugars by about 10%. The IS sugar was the sample with poorest flowability, based on a Hausner ratio equal to 1.33. Compared to the others, this sugar variety presented a higher bed porosity and a smaller tapped density. In spite of its irregular shaped particles, the RS sugar compacted easily and achieved the same tapped density of samples with well-defined granular particles. A slower compaction kinetic was observed for the coarser sugar—VS ($D_{50} = 696 \mu\text{m}$). Finally, the static and dynamic repose angles increased from the granular (GS and VS) to the powdery sugars (RS and IS), but for the latter ones the values were not significantly different in spite of the mean sizes decreased from 206 μm in the RS sugar to 77 μm in the IS. A more relevant difference was observed in the dynamic repose angle for the powdery sugar and the difference between the dynamic and static repose angles increased as the mean particle size was reduced. According to their repose angles, the flowability would be classified as good (GS and VS) or fair (RS and IS). The densification dynamics of all sugars were fitted with good accuracy to an empirical equation.

Practical applications

Powder flow is influenced by the presence of fines contained in the bulk powder, by the particles' size distribution and by their morphological characteristics. To meet the different needs of customers, a number of sugar varieties are marketed, including the granulated white sugar, the brown raw sugars and the powdery refined sugars. Every sugar variety behaves differently in operations such as packing, transport, and discharge from silos and hoppers. As sugar manufacturing plants deal with materials in a broad range of sizes and shapes, data of basic flow indicators for different sugar varieties obtained under standardized and comparable conditions may be useful for predicting the performance of a processing plant and monitoring the product quality attributes and for designing equipment as well.

1 | INTRODUCTION

More than half of all the sugar commercialized in the world is extracted from sugarcane and produced in Brazil (MAPA, 2016). According to the Brazilian Sugarcane Industry Association UNICA (2016), approximately 25 metric tons of sugar were exported in 2014/2015, an amount that

corresponds to 68% of the national production. To meet the different needs of customers, a number of sugar varieties are marketed, including the granulated white sugar, the brown raw sugars and the powdery refined sugars. They are composed by sucrose at purity levels over 98% (Mathlouthi & Reiser, 1995). In processing, these varieties undergo similar operations for sucrose extraction, however, the white sugars are

submitted to additional stages of blanching and purification and the refined sugars are milled and sieved to yield powders under specified size ranges (Mathlouthi & Reiser, 1995). Therefore, the size distributions of sugar powders can range from very fine powders (such as icing sugar) to coarse granules (such as the crystal-like sugars). Powder flow can be largely influenced by the presence of fines contained in the bulk powder, by the particles' shape and also by their surface characteristics (Bodhmaghe, 2006). Therefore, every sugar variety behaves differently in operations such as packing, transport, and discharge from silos and hoppers. As sugar manufacturing plants deal with materials in a broad range of sizes and shapes, reliable data on basic flow properties of distinct sugar varieties may be useful for designing equipment, predicting the performance of processing plant and monitoring the quality attributes of the products. Some easily measurable bulk properties, such as the repose angles (AR) and the Hausner ratio (HR) are considered good indicators of powder flowability under low compaction stresses, as they provide meaningful, easily interpretable and easily usable results that may be used to improve an industrial process. While the static repose angle gives a measure of powder cohesion when the powders are submitted to a gravitational flow to form a heap, the dynamic repose angle (AR_d) gives an equivalent measure for a powder submitted to a rotational movement thus simulating dynamic conditions that are common in industrial processes (Bodhmaghe, 2006; Lumay et al., 2012).

This study was motivated by the observation that, despite the economical relevance of this commodity, basic data on powder characterization for different sugar varieties are scarce in the literature, as well as studies on powder flowability of samples with multimodal particle size distributions. We focused in obtaining flowability indexes based on Hausner ratio and repose angle and on analysis of compaction behavior because these are simple and robust methods that can be used to monitor the flow behavior of production units on daily basis. The shearing tests, although useful to predict the flow behavior of consolidated powders, depend on the determination of flow functions, which is time consuming and of difficult reproducibility (Fitzpatrick, Barringer, & Iqbal, 2004).

It is common knowledge that the compaction behavior and repose angle of bulk powders are quite sensitive to the particle size distribution of samples. A review of literature showed that measurements of repose angles and bulk densities have already been investigated for monodispersed samples (Lumay et al., 2012), for powders with narrow particle size distributions (Pohlman, Roberts, & Gonser, 2012), and for samples prepared by adding different percentages of fines to a base sample (Geldart, Abdullah, Hassanpour, Nwoke, & Wouters, 2006). These particle size distribution models, however, are not representative of the typical samples handled in manufacturing plants. In the consulted literature, we found no study investigating whether these methods are really effective in predicting differences in the flowability behavior of sucrose powders which underwent to different processing steps and have multimodal size distributions and wide size spans.

The purpose of this study was to obtain some flowability indicators for commercial sugar powders, namely raw type sugar (also known as VHP sugar and denominated VS in this text), a white crystal-like sugar (GS), refined sugar (RS), and icing sugar (IS). These materials were

characterized according to their morphology and particle size distribution, and the measured properties were the static and dynamic repose angles, the aerated and tapped bulk densities, as well as the Hausner ratio. The densification dynamics was obtained by tapping and the data were fitted to an empirical equation to describe the compaction behavior of the sugars.

2 | EXPERIMENTAL

2.1 | Materials

The VS sugar samples used in the experiments were obtained from a processing plant located in Pradópolis –SP, Brazil (Usina São Martinho). The white crystal-like sugar (GS) was from Colombo brand, while refined sugar (RS) and icing sugar (IS) were got from União. All of them were acquired from local supermarkets in São Carlos–SP, Brazil.

2.2 | Characterization

The sugar samples were characterized according to the methods described in the following section.

2.2.1 | Particle size measurements

Size distribution of IS sugar was obtained by laser light scattering methodology in an MAF 5001 analyzer from Malvern Instruments (Malvern, Malvern Hills). The samples were dispersed in a liquid phase (ethanol). VS, GS, and RS sugars were sieved in a set of sieves from 18 to 70 mesh (for VS and GS) and from 18 to 170 mesh for the RS samples, using an AS 200 shaker from Retsch (Haan, Mettmann). The sieves followed the standard ISO 3310 manufacturing protocol and each sample was sieved twice for 10 min under a vibration amplitude of 3 mm. All the size distribution analysis was performed in triplicate.

2.2.2 | Electronic scanning microscope

Samples were placed on discs (first wrapped with carbon tape) prior to each analysis. These discs were taken to gold coating for 90 s on Sputter Coater SOD004 from Balzers (Balzers, Oberland) and then micrographs of the sugar particles were obtained from an MEV Inspect S50 from FEI (Hillsboro, Oregon) using an Everhart-Thornley detector (ETD).

2.2.3 | Moisture content

The moisture content was determined in triplicates, according to the methodology recommended by AOAC (2000). Samples of 5 g were dried at 100 °C in a 315SE oven from FANEM (São Paulo, Brazil) until the differences in weight were lower than 2 mg after two successive measurements. The saturation stage was calculated by:

$$S(\%) = \frac{V_w}{V_t - V_s} \times 100 \quad (1)$$

where V_w is the volume of liquid water in the packed-bed; V_t is the total volume of the compacted dry bed and V_s is the volume estimated from the particle density, which was measured using a gas pycnometer

Accupyc II 1340, from Micromeritics (Gwinnett County, Georgia) using a 10 cm³ volume chamber.

2.3 | Flow properties

2.3.1 | Repose angles

The experimental set-ups and procedures for measuring both static and dynamic repose angles are described in the following section.

Static repose angle (AR)

The experimental set up consisted of a funnel and a glass plate. A standardized protocol was adopted to pour the sugars because the repose angle is very sensitive to the method used to create the heap (Lumay et al., 2012). A mass of 200 g was used for the coarse sugars (VS and GS) and of 100 g for the refined sugars (RS and IS). The funnel bottom was initially blocked, and after it was filled with powder, the discharge was released over the glass plate of 10.6 cm in diameter. The distance from the funnel outlet orifice to the base surface was 10 cm. The IS sugar samples were sieved before filling up the funnel to avoid agglomerates. Three tests were performed for each sample. The powder piles were photographed using a DSC TX20 digital camera from Sony (Minato, Tokyo) and the angle of the conical heap formed was determined by:

$$AR(^{\circ}) = \tan^{-1} \frac{h}{R} \quad (2)$$

where R is the radius of the base surface and h is the pile height.

Dynamic repose angle (AR_d)

The dynamic repose angle was measured using a rotating drum. A horizontal acrylic cylinder of diameter $D = 12$ cm and length $L = 6$ cm was half filled with the powder. The cylinder rotates around its axis at a constant velocity resulting in the powder flow. The flow dynamics of granular powders in rotating drums is a function of the rotating velocity and according to Lumay et al. (2012), for measuring the dynamic repose angle of non-cohesive granular material, the rotating drum speed range should be selected in order to obtain a continuous flow regime with flat interface. For the GS and RS sugars, the desired regime was observed at a rotating velocity equal to 2 rpm, however, for the VS sugar, it was observed only at 7 rpm. This ensured that measurements were performed under identical dynamic regimes and the results are comparable. The images were recorded using a DSC TX20 digital camera from Sony (Minato, Tokyo). From the videos, five frames were randomly selected and used for measuring the repose angle. These images were processed using the software Image Pro-Plus 6.0 from Media Cybernetics (Rockville, Maryland) and for each condition, the angle formed by the air-powder interface with the horizontal line was determined.

2.3.2 | Loose and tapped bulk densities and Hausner ratio

The loose bulk density was measured by pouring the samples into a 250 ml glass measuring cylinder, following the protocol described in Section 2.3.1, as the packing patterns are highly dependent on the method used for packing (Zou & Yu, 1996). The tapped bulk density

TABLE 1 Classification of solid particle flowability according to classical methods: Static repose angle and Hausner ratio

Flowability classification	Repose angle (°)	Hausner ratio
Excellent	25–30	1.00–1.11
Good	31–35	1.12–1.18
Fair	36–40	1.19–1.25
Passable	41–45	1.26–1.34
Poor	46–55	1.35–1.45
Very poor	56–65	1.46–1.59
Very very poor	>66	>1.60

Source: European Pharmacopoeia, 2010

was determined using a Tap2S automatic device from Logan Instruments (Somerset, New Jersey) until a constant volume, according tapping “method I” from U.S. Pharmacopeia (2000). All density determination tests were performed twice and the Hausner ratio and the bed porosity were calculated from Equations 3 and 4, respectively:

$$HR = \frac{\rho_{tb}}{\rho_{lb}} \quad (3)$$

$$\varepsilon = 1 - \frac{\rho_{lb}}{\rho_p} \quad (4)$$

where ρ_{tb} is the tapped bulk density, ρ_{lb} is the loose bulk density and ρ_p is the sugar particle density obtained by gas picnometry.

A classification of powder flowability according to the Hausner ratio and the static repose angle is presented in Table 1.

2.4 | Compaction behavior

The samples of aerated powders were compacted using the methodology described in Section 2.3.2. The powder volumes were recorded as a function of the number of taps (N), until $N = 1,000$ because no variation in the bulk powder volume was observed for $N > 1,000$. The volume of pores was obtained from compaction and the specific pore volume was plotted as a function of the number of taps to evaluate the densification dynamics of sugar powders. The empirical equation developed by Mallof, Amorós, Orts, and Llorens (2008) was fitted to the empirical data:

$$1 - \frac{\xi}{\xi_{\max}} = f e^{\left(-\frac{N}{N_1}\right)} + (1-f) e^{\left(-\frac{N}{N_2}\right)} \quad (5)$$

where f , N_1 , N_2 , and $\xi_{\max}$ are the fitted kinetic parameters, ξ is the degree of advance in the powder densification process, defined as:

$$\xi = \frac{V_{p0} - V_p}{V_{p0}} \quad (6)$$

where V_{p0} and V_p are the initial pore volume (when the sample is aerated) and the pore volume in a specific tap, respectively.

The parameter f represents the densification fraction produced by the elimination of arches or large pores; N_1 represents the number of taps at which 63% of the large pores or voids are eliminated; N_2 is the

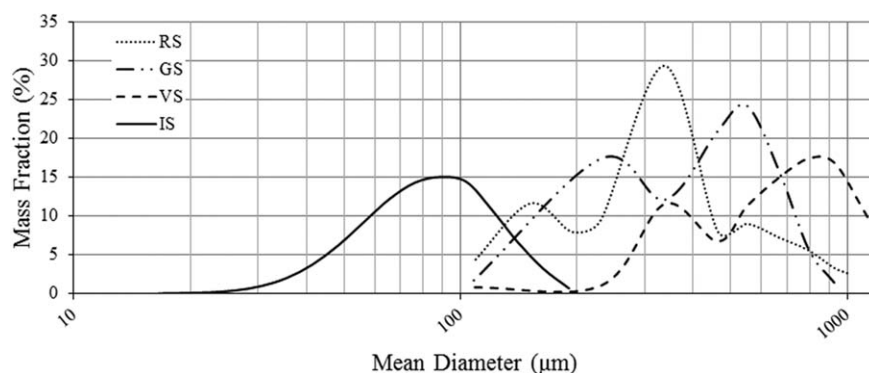


FIGURE 1 Size distribution for sugars

number of taps at which 63% of the small pores are eliminated and $\xi_{\max}$ is the maximum degree of advance reached by the powder.

2.5 | Statistical analysis

The one-way analysis of variance (ANOVA) based on the Tukey test was used to verify whether there are any statistically significant differences ($p < .05$) between the means of different groups of measured flow properties performed by Excel software from Microsoft (Redmond, Washington). The data are reported as mean values and their standard deviation values.

3 | RESULTS AND DISCUSSIONS

The results of powder characterization, densification analysis, and flowability indicators are presented in the following sections.

3.1 | Characterization

3.1.1 | Particle size distribution, physical properties, and morphology

The particle size distributions for GS, VS, RS and IS sugars are shown in Figure 1. Table 2 shows the values of D_{10} , D_{50} , D_{90} , the size span, the moisture content and the saturation state of the sugars.

GS and VS sugars are formed of coarse granules with bimodal particle size and asymmetric distributions, RS sugar shows particles from approximately 100 to 1,000 μm and the wider size span from all the samples. IS variety has 100% of particles with sizes under 200 μm and Gaussian asymmetric particle size distribution.

The presence of fine particles in bulk powders increases the specific superficial area and enhances the action of interparticle attractive forces such as the Van der Waals forces, which may lead to a cohesive behavior and poor flowability (Castellanos, 2005). Chitprasert, Chedchant, Wanchaitanawong, and Poovarodom (2006) suggested that particles under 425 μm in sugar powders may be a potential source of flowability issues. It can be noted in Table 2 that, besides the IS sugar, the RS sugar also has a significant mass fraction of fine particles, with $D_{10} = 123 \mu\text{m}$.

According to Table 2, the moisture content of samples is low, as the maximum value observed for the RS variety was only 0.31%. The water saturation level in the powders ranged from 0.07 to 0.79%. According to Althaus and Windhab (2012), water saturation levels up to 50% characterize a pendular-state stress condition and at such low moisture levels, the development of capillary forces is not significant and powder cohesion is not affected by the presence of interparticle liquid bridges.

The morphology of sugars can be observed in the MEV micrographs shown in Figure 2a,b.

It can be observed that the VS and GS sugars are formed by cubic crystals, the IS sugar is formed by agglomerated particles and the RS sugar by rounded particles with a wide size distribution. It can be observed in the 1,000x magnified micrographs (Figure 2b) that the GS and VS sugar crystals have fine particles adhered to their surfaces, which causes some roughness to them. The IS sugar particles agglomerate due to the action of attractive interparticle forces, which are relevant in this size range and increase powder cohesiveness. It is worth noting that Figure 1 shows the size distribution of the primary particles of IS sugar and not the agglomerates.

TABLE 2 Sizes and moisture content of sugar powders

	D_{10} (μm)	D_{50} (μm)	D_{90} (μm)	Size span	Moisture content (% w.b.)	S (%)
VS	306 ± 6^a	696 ± 7^a	1062 ± 6^a	1.09 ± 0.03^a	0.08 ± 0.03^{bc}	0.28 ± 0.01^b
GS	276.0 ± 0.7^b	537 ± 2^b	772 ± 3^b	0.92 ± 0.00^b	0.03 ± 0.02^c	0.08 ± 0.02^b
RS	123 ± 5^c	286 ± 5^c	704 ± 16^c	2.03 ± 0.04^c	0.31 ± 0.02^a	1.11 ± 0.27^a
IS	44.6 ± 0.4^d	76.9 ± 0.7^d	122 ± 1^d	1.00 ± 0.00^d	0.21 ± 0.07^{ab}	0.52 ± 0.00^b

Values in the same column with different letters are significantly different at a .05 significance level.

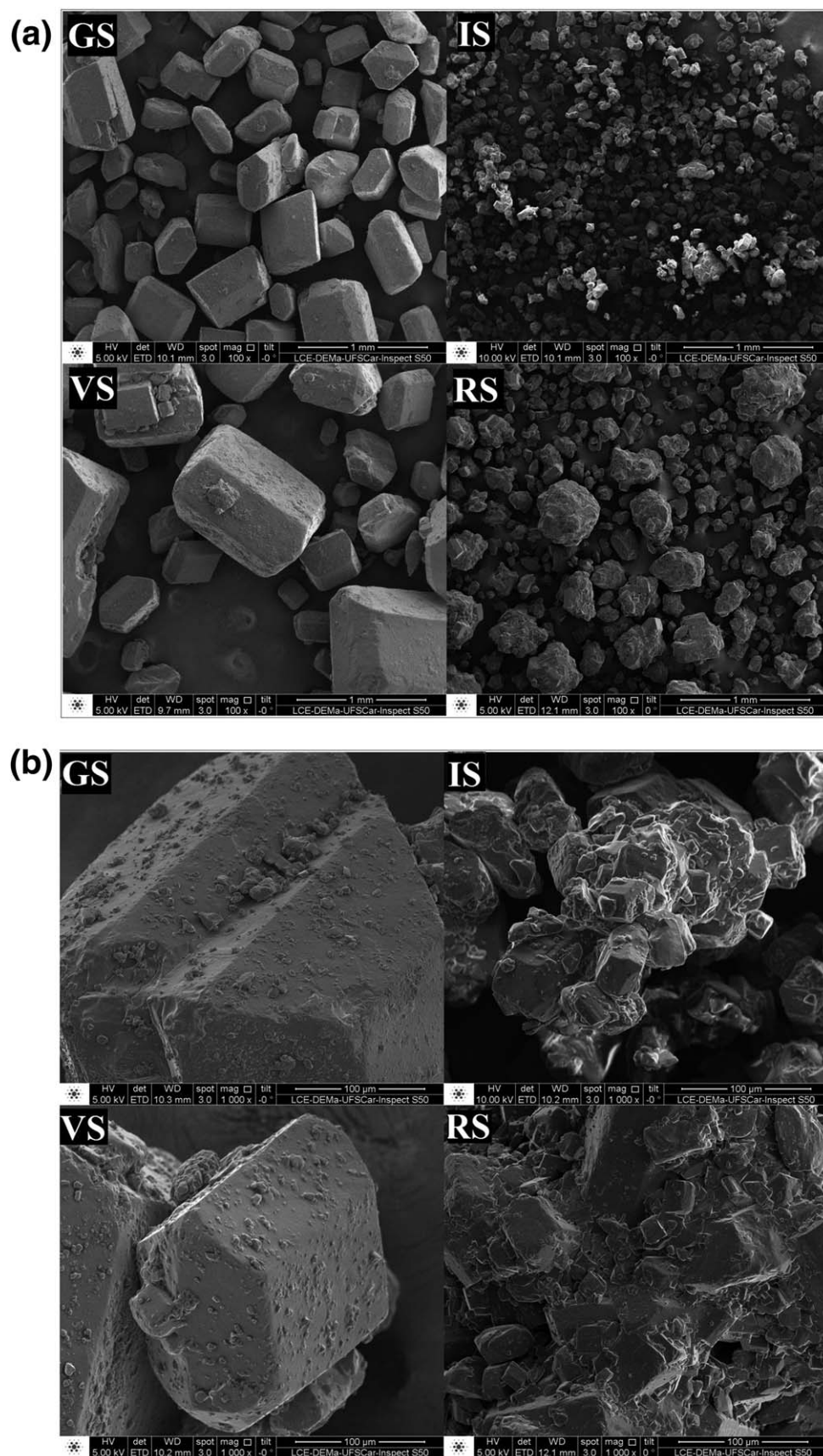


FIGURE 2 Micrographs of sugar particles. (a) 100 $\times$ magnification; (b) 1,000 $\times$ magnification

TABLE 3 Particle, loose and tapped bulk densities for different types of sugar and loose porosity of their beds

	ρ_p (g/cm ³)	ρ_{lb} (g/cm ³)	ρ_{tb} (g/cm ³)	Bed porosity (ϵ)
VS	1.5885 $\pm$ 0.0004	0.866 $\pm$ 0.005 ^a	0.962 $\pm$ 0.000 ^b	0.455 $\pm$ 0.003 ^a
GS	1.5763 $\pm$ 0.0002	0.897 $\pm$ 0.006 ^b	0.995 $\pm$ 0.007 ^a	0.431 $\pm$ 0.004 ^b
RS	1.5777 $\pm$ 0.0006	0.777 $\pm$ 0.004 ^c	0.949 $\pm$ 0.005 ^b	0.508 $\pm$ 0.002 ^c
IS	1.5998 $\pm$ 0.0008	0.630 $\pm$ 0.005 ^d	0.840 $\pm$ 0.009 ^c	0.607 $\pm$ 0.003 ^d

Values in the same column with different letters are significantly different at .05 significance level.

3.2 | Loose and tapped bulk densities

The values of loose (ρ_{lb}) and tapped (ρ_{tb}) bulk densities are shown in Table 3, as well as the packed-beds' mean porosity. The tapped density was obtained from the volume measured at full compaction, $N = 1,000$.

The particle densities of all powders (ρ_p) are similar because sucrose has a compact crystalline structure with few inner voidages. The measured values agree with that reported by IFA (2016) for crystalline sucrose, equal to 1.5737 g/cm³. The ρ_{lb} values are close for sugars GS and VS indicating that they tend to settle similarly as poured into a container. This result may be attributed to the similar shape of their particles, which are both regular and uniform and to the fact that their size distributions, although statistically different, have common features such as the bimodal pattern and the significant fraction of coarse particles. On the other hand, RS and IS sugars presented decreasing values of ρ_{lb} . The presence of agglomerates in IS sugar leads to a low ρ_{lb} , as the agglomerates contribute to forming large voids in the packed-bed, therefore, a higher volume is necessary to accommodate a given mass of particles as they are settled. RS sugar shows an intermediate value of ρ_{lb} , which is a consistent result, as D_{50} of this powder is smaller than D_{50} of GS and VS sugars, and larger than the IS sugar.

The kinetic curves showing the pore size reduction along the compaction are presented in Figure 3.

It can be seen that the pore volumes in all samples are rapidly eliminated in the beginning of the densification process. Table 4 shows the total volume reduction reached in the compaction of each sample and the respective values of N_{50} , which is the number of taps required to reduce the initial bulk volume by 50%.

As seen in Table 4, for IS sugar, 11 taps were necessary to reduce the initial bulk volume by 50%. For GS and RS sugars, an equivalent volume reduction was observed after eight taps, and for VS sample after 33 taps. The VS sugar is the sample with higher values of D_{50} and D_{90} and, as it would be expected, coarser particles are not capable of filling in easily the interparticle voids in the packing structure, which explains the slower compaction kinetics of this sugar. After $N = 1,000$ taps, the GS and VS sugars had their initial volumes reduced only by 9 and 10%, respectively. The compaction was greater for IS and RS sugars, whose initial volumes were reduced by 25% and 18%, respectively. The more intense compaction of IS sugar is explained because this sample has a large amount of pores ($\epsilon = 0.607$) and higher loose bulk density. Szulc and Lenart (2010) investigated the compaction of ground sugar particles ($D_{50} = 208 \mu\text{m}$) by tapping and reported a compaction of 30% of the bulk volume.

It can be observed in Figure 3 that the compaction behavior is similar for VS, GS, and RS sugars. In fact, except for the initial point, the solid lines of RS and VS sugars are practically overlapped in Figure 3. The similarity between VS and GS sugars should be expected as these samples are much alike regarding their particle size distributions and shape. However, the RS sugar is a sample with D_{50} , shape and a size distribution quite different from the VS and GS sugars. This result illustrates the complexity involved in predicting the powder behavior in bulk solids handling, particularly for powders with a wide-ranged size distribution. The reduction in the pore volumes in the beginning of the compaction kinetics is more intense for the IS sugar in comparison to the others, although the global qualitative behavior of the curve is similar. Owing to the agglomerated particles, the powder packed-bed has a larger volume of initial pores to be eliminated. Under tapping, the agglomerates are broken and the fine particles may fill in these voids

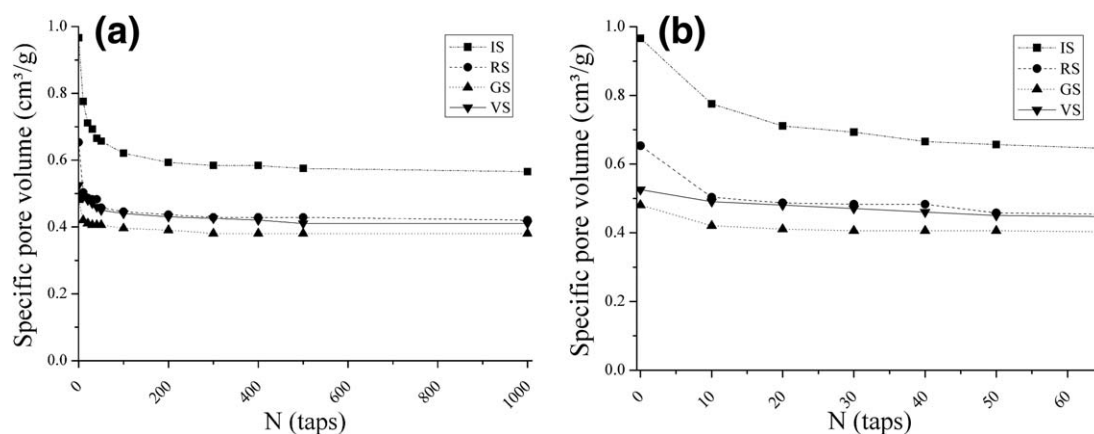


FIGURE 3 (a) Pore size reduction as a function of the number of taps; (b) Pore size reduction at interval $0 < N < 60$

TABLE 4 Bed compaction by tapping method ($N = 1,000$) for the sugar varieties

	% Change in bed volume by tapping	N_{50}
VS	10.0 ± 0.6^c	36 ± 4^a
GS	9.0 ± 0.1^c	9 ± 0^b
RS	18.1 ± 0.8^b	8 ± 1^b
IS	25 ± 1^a	15 ± 3^b

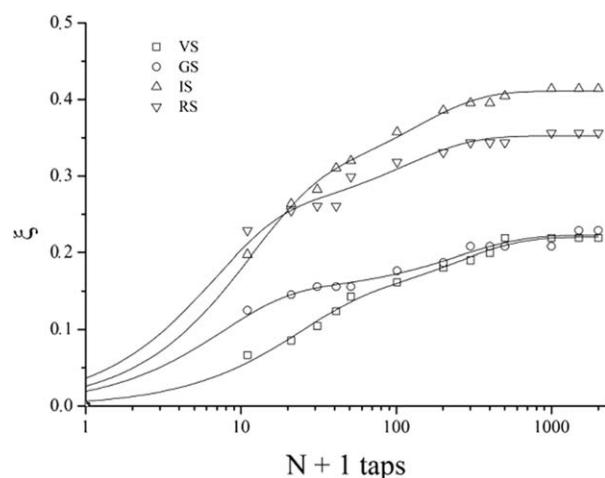
Values in the same column with different letters are significantly different at a .05 significance level.

easily, therefore the curve shows a rapid initial decrease in the pore volume. After $N = 100$, the volume changes at lower rates and the curve reproduces the qualitative behavior observed for the other sugars.

In Figure 4, the fitting of Equation 5 (solid line) to the experimental data (symbols) can be compared.

It is worth noting that Equation 5 was originally proposed based on experimental data obtained for samples of spherical quartz particles, with D_{50} from 32.4 to 409.8 μm and monomodal size distributions. In spite of the differences in sizes and shape of the particles, the equation fitted well our experimental data, as can be observed in Figure 4.

The fitted parameters, f , N_1 , N_2 , and ξ_{max} as well as the statistic parameters of the fit are shown in Table 5. According to Mallol et al. (2008), the increase in ξ with N is due to a decrease in the volume of large pores and small pores. Large pores are those formed by voids and cavities larger than the particle size, while small pores are those of a size similar to the particle size. It can be seen in Table 5 that the parameters f and N_1 of IS, GS, and RS sugars are very similar, a result that reflects the compaction kinetics behavior observed in Figure 4. The major difference was observed in parameter $N_1 = 19$ fitted for GS sugar, which is significantly higher than those fitted for the other samples. N_1 indicates the number of taps in which 63.5% of the large pores is eliminated. The higher value observed for VS sugar is consistent with the results presented in Figure 3, as it shows that the initial compaction

**FIGURE 4** Fitting of Equation 5 to the sugar powders compaction process**TABLE 5** Kinetic and statistic parameters fitted to Equation 5

Sugar type	Fitting parameters				Statistic parameters for fitted equations		
	f	N_1	N_2	ξ_{max}	SQR	ERM (%)	R^2
IS	0.67	9	135	0.41	1.040×10^{-3}	1.23	0.9922
VS	0.58	19	221	0.22	3.786×10^{-4}	3.66	0.9919
GS	0.65	6	298	0.23	8.097×10^{-4}	2.06	0.9767
RS	0.67	4	110	0.36	2.440×10^{-3}	2.02	0.9724

of this sugar is slightly slower than the observed for the others. This behavior suggests that eliminating small pores is more difficult in this sample due to the presence of coarse particles. The fitted values of ξ_{max} were higher for the IS and RS sugars, corroborating that the fine powders are easily compacted. The statistical parameters show small values of SQR (lower than 10^{-2}), ERM values under 5% and R -square values close to 1, confirming the good fitting.

3.3 | Flow properties

Results found for the Hausner ratio, static and dynamic repose angles are presented in Table 6. The HR values were obtained from the measured values of loose and compacted bulk densities shown in Table 3.

Based on the HR values, none of the sugars is classified as a cohesive powder, as the highest value, observed for the IS sugar, was equal to 1.33. According to Table 1, this HR is characteristic of passable flowability, however, this powder clearly presented a more difficult handling behavior as compared to the others. This result is consistent with the size distribution of this powder ($D_{50} = 77 \mu\text{m}$ and $D_{90} = 122 \mu\text{m}$). For ground sugar of $D_{50} = 200 \mu\text{m}$, Szulc and Lenart (2010) reported on HR values of about 1.36. The particle sizes are quite different, but it is known that the flow properties do not depend only on the particle size, but also on additional physical features of the particles, such as shape and surface properties. The RS sugar flowability was classified as fair, and the GS and VS sugars flowability as excellent.

Table 6 also shows that the static repose angles of GS and VS sugars are close to 30° , which characterizes a free-flowing material. The angles measured for the RS and IS sugars were also similar, equal to $38 \pm 1^\circ$ and $40 \pm 1^\circ$, respectively. Based on these values, the flowability of these powders was categorized as fair. As observed in the

TABLE 6 Flow properties of different types of sugar

Sugar	VS	GS	RS	IS
HR	1.11 ± 0.01^c	1.10 ± 0.00^c	1.21 ± 0.01^b	1.33 ± 0.04^a
AR ($^\circ$)	32.2 ± 0.9^b	30.9 ± 0.5^b	38.4 ± 1.3^a	40.3 ± 0.6^a
AR _d ($^\circ$)	40 ± 1^b	41 ± 1^b	50 ± 4^a	55 ± 5^a
Flowability ^a	Excellent	Excellent	Fair	Passable

Values in the same line with different letters are significantly different at .05 significance level.

^aAccording to HR.

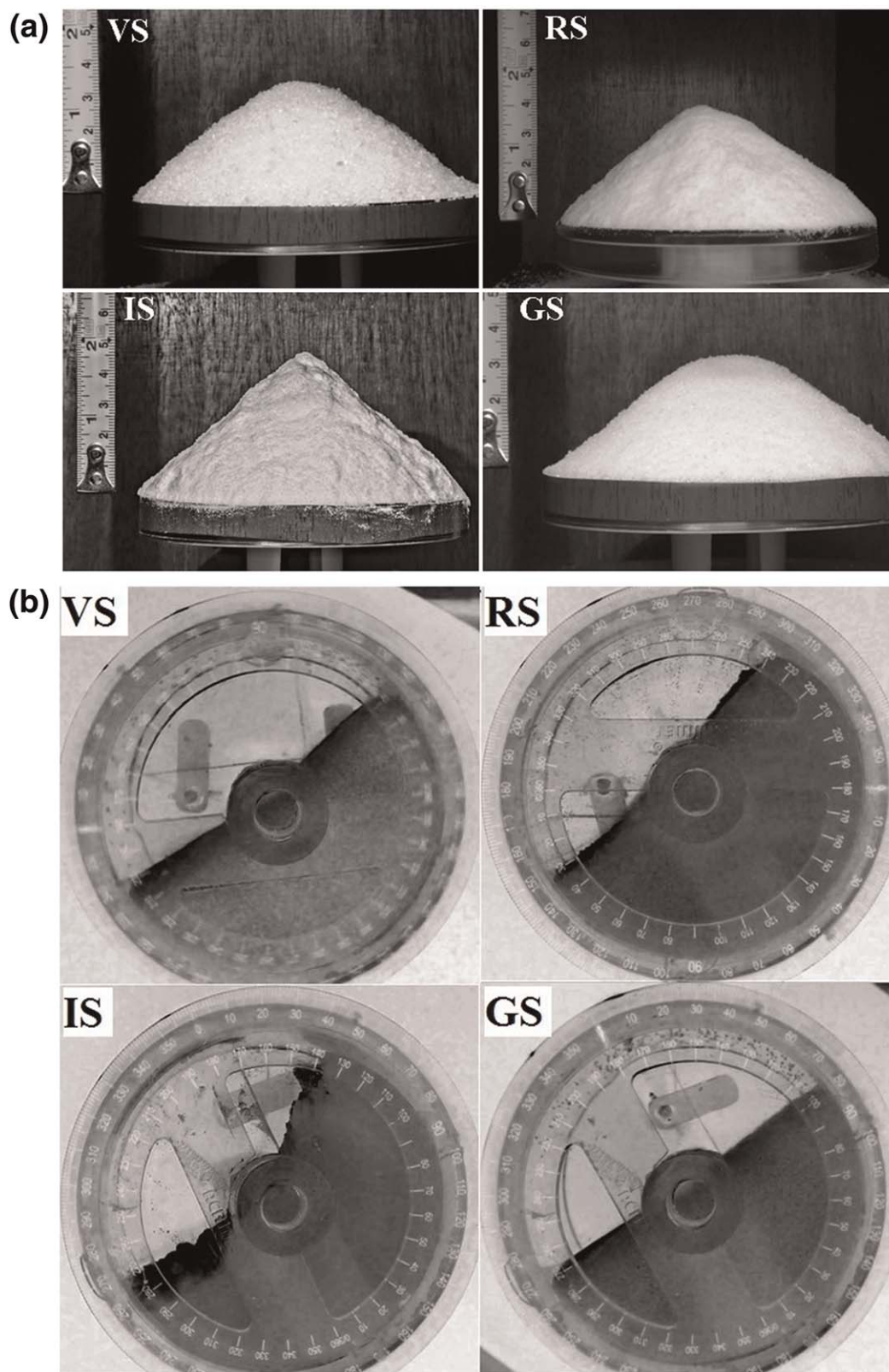


FIGURE 5 (a) Sugar piles observed during measurement of the static repose angle. (b) Powder flow in the rotating drum observed in the measurement of the dynamic repose angle

analysis of compaction kinetics, the close values of AR for GS and VS sugars are easily explained by the similarity of the shape and size distributions of these sugars. The static repose angle however, was not

sensitive to the different size distributions of RS and IS sugars, as the values of the AR of these sugars were not significantly different if the standard deviations are considered.

For the coarse sugars, VS and GS, the AR_d values were close to 40° and for RS and IS sugars, the values were equal to 50° and 55° , respectively. Although the value of AR_d for the IS sugar is consistently higher than that of the RS sugar, they are not significantly different if the standard deviations are considered. This is evidence that, as observed for the static repose angle, the technique was not sensitive to the significant differences in these powders' properties. Lumay et al. (2012) measured the dynamic repose angles of cohesive and non-cohesive granular materials in a rotating drum and observed that they increased with powder cohesion. According to the authors, the macroscopic properties of a powder are influenced by particles of diameters under $50\text{ }\mu\text{m}$, which are mainly related to the cohesive forces between the particles. From the samples tested, the IS sugar is the only powder with D_{10} under $50\text{ }\mu\text{m}$, which contributes to increasing the AR and AR_d .

To illustrate the sugar piles formed when measuring the static repose angles and the flow patterns observed in the measurement of dynamic repose angles, some photographs representative of tested conditions are depicted in Figure 5.

It can be observed in Figure 5a that the granular sugars GS and VS form conical heaps with rounded peaks. The heaps formed by RS and VS sugars show a slightly different pattern, with pointed peaks and irregular surfaces. According to Lumay et al. (2012), the heap shape strongly depends on the grain properties and a cohesive granular material gives a higher value of the repose angle. Fine-powdered sugars form irregular heaps due to cohesiveness induced by the small size of the particles, as the cohesive forces rule the macroscopic properties of the bulk solids. On the contrary, in non-cohesive granular sugars, the macroscopic properties are governed by the geometry of the grains and by their surface properties, which are related to the steric repulsion and to the friction forces. In Figure 5b, it can be observed the regular and flat interfaces achieved in the rotating drum for GS, VS, and RS sugars. They reproduce the classical pattern of non-cohesive granular materials rotating under a continuous flow and low Froude numbers (Lumay et al., 2012). The IS sugar is a more cohesive powder and the continuous regime was not observed for this sample. The patterns observed for the IS sugar in Figure 5b agrees with that reported by Lumay et al. (2012) for cohesive powders.

Qualitatively, the AR and AR_d presented similar variations regarding the powder size, with higher values for the finer samples. The powders in a rotating drum are submitted to gravitational and centrifuge forces opposing the friction and cohesive forces, therefore, the dynamic angles are always higher than the static ones.

4 | CONCLUSIONS

The densification dynamics and flow properties of powdery to granular varieties of commercial sugars were obtained. The results showed that these properties are linked to the powder size distribution, whose D_{50} ranged from 77 to $696\text{ }\mu\text{m}$, but the correlation is not always direct. Moisture content, which is often responsible for enhancing the cohesion in bulk solids, had no influence on the flowability behavior of tested sugars, as the saturation levels of all sugar varieties were lower than the value that characterizes a funicular state. The higher moisture

content of the RS sugar revealed some irregular and rough agglomerates, while the samples with lower moisture content were composed of flat and well-defined crystals.

It was observed a reduction of 10% in the pore volume of the granular sugars (GS and VS) by tapping, of about 18% for the RS and 25% for the IS sample, which exhibited a more difficult handling as compared to the other sugars. None of the sugar assemblies could be classified as cohesive, but the IS variety agglomerated into clusters with intraparticle voidage, resulting in beds of higher porosity and smaller tapped density as compared to the other sugar varieties. In spite of its irregular shaped particles, the RS sugar compacted easily and achieved the same tapped density of samples with well-defined granular particles. A slower compaction kinetic was observed for the coarser sugar – VS ($D_{50} = 696\text{ }\mu\text{m}$). Finally, the static and dynamic repose angles increased from the granular (GS and VS) to the powdery sugars (RS and IS), but for the latter ones the values were not significantly different in spite of the mean sizes decreased from $206\text{ }\mu\text{m}$ in the RS sugar to $77\text{ }\mu\text{m}$ in the IS. A more relevant difference was observed in the dynamic repose angle for the powdery sugar and the difference between the dynamic and static repose angles increased as the mean particle size was reduced.

NOMENCLATURE

AR and AR_d	Static and dynamic repose angles ($^\circ$)
D_{10}	Particle diameter at 10% of cumulative size distribution (L)
D_{50}	Particle diameter at 50% of cumulative size distribution (L)
D_{90}	Particle diameter at 90% of cumulative size distribution (L)
f	Fitted parameter in Equation 5
h	Pile height in measuring repose angle (L)
HR	Hausner ratio
N	Number of taps
N_1 and N_2	Fitted parameters in Equation 5
N_{50}	Number of taps to reduce bulk powder volume in 50%
R	Radius of the base surface in measuring repose angle (L)
S	Saturation level of water (%)
V_p and V_{p0}	Pore volume and initial pore volume in loose packed-bed (L^3)
V_s	Estimated volume for the particle (L^3)
V_t	Total volume of the compacted dry bed (L^3)
V_w	Volume of water in the samples (L^3)

Greek symbols

ε	Bed porosity
ρ_{lb}	Loose bulk density (M L^{-3})
ρ_p	Particle density (M L^{-3})
ρ_{tb}	Tapped bulk density (M L^{-3})
ξ and $\xi_{\max}$	Degree of advance defined in Equation 6 and maximum degree of advance fitted in Equation 5

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